

Rahul Son Boro

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EDUCATION

Central Institute of Technology Kokrajhar

B.Tech in Computer Science and Engineering — CGPA: **7.59/10.0**

Assam, India

Jul 2023 – Jun 2026

Central Institute of Technology Kokrajhar

Diploma in Computer Science and Engineering — CGPA: **7.46/10.0**

Assam, India

Jul 2020 – Jun 2023

EXPERIENCE

Work Based Learning Intern

NIELIT Kokrajhar EC

Feb 2026 – Jul 2026

Assam, India

- **Research Aggregator Agent:** Built an AI search engine querying **Semantic Scholar**, **ArXiv**, and **GitHub APIs** in parallel; LLM generates grounded, hallucination-free 2-line summaries of papers.
- Cut research discovery time from hours to <10 sec via card-based UI displaying title, citations, PDF and summary; planned **“Chat with Paper”** feature.

Software Engineering Intern

APT Software Avenues Pvt. Ltd.

May 2025 – Jul 2025

Kolkata, India

- **Image/Pdf - to - DOCX:** Two-stage converter using **PaddleOCR (PP-StructureV3)** → Markdown → **Editable-docx(Post-processor)**; reconstructs tables, syntax-highlighted code (**Pygments**), and Mermaid diagrams.
- **SkillBuilder AI:** Serverless **Telegram** bot with **n8n**, **Google Gemini (LangChain)**, and **YouTube API** for adaptive micro-learning through Youtube videos; auto-generates daily roadmaps and quizzes.

PROJECTS

Personalized UI Aesthetic Score Prediction: | *EfficientNet-B0, FedAvg, Ollama*

Jan 2026 – May 2026

- Curated a 100-UI paired dataset (desktop+mobile, 10 domains) rated by 100 users with demographics; trained a FedAvg (EfficientNet-B0) model with per-user rating centering to predict **individualized** 1-10 scores (MSE 0.0534).
- Engineered user-adaptive normalization (centering + min-max scaling) enabling a single global model to serve 5 demographic groups; deployed via Gradio and compared against zero-shot Qwen-VL-235B to validate data-driven personalization.

Dual-Purpose Mall Monitoring: | *VGG16, YOLOv8, MediaPipe, SlowFast*

Jan 2025 – May 2025

- Self-curated video dataset from mall surveillance; built real-time threat detection pipeline using **YOLOv8** (object detection), **MediaPipe Pose** (skeletal keypoints), and **SlowFast Network** (action recognition) to identify theft/hostile activity.
- Integrated **VGG16-based CNN** for facial expression analysis (FER2013, 80% accuracy); dual outputs for customer satisfaction + anomaly alerts.

Assamese Movie Recommender: | *TF-IDF, Cosine Similarity, Pandas, NumPy*

Jan 2023 – May 2023

- Created dataset of 1,119 Assamese films; built content-based recommender using **TF-IDF vectorization** + **Cosine Similarity** on metadata (genres, cast, director, keywords); integrated **DiffLib** for fuzzy input matching.
- Overcame cold-start problem for niche regional cinema; delivered personalized suggestions via efficient NumPy/Pandas similarity computations.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (Postgres), JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, NextJS, Express.js, Node.js, MongoDB, Redux, Tailwindcss, WordPress, Material-UI, FastAPI

Cloud & DevOps/Version Control: Git/Github, Linux/UNIX, Docker, AWS, Google Cloud Platform

Security & Tools: Wireshark, PortSwigger

Specializations: Web Development, ML & DL, Computer Vision, AI, AI Agents(LangChain & LangGraph)

LEADERSHIP & ACTIVITIES

Leadership: Technical Coordinator at Coding Club CITK (2022-2023)